Signal State Dictionary — Options Quant Engine

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Overview

This document catalogues every symbolic state label, regime label, and classification term used by the Options Quant Engine. Each entry specifies the term's origin module, decision rule with exact thresholds, upstream inputs, and downstream interpretation.

Total terms documented: 91
Categories: 28 + 3 composite metrics
Layers traversed: Market Data → Analytics → Risk/Overlay → Signal Engine → Trade Decision

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State Dependency Diagram

The diagram below shows how raw market data flows through successive classification layers to produce a final trade decision. Each box represents a classification function; arrows show data dependencies.

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                              MARKET DATA LAYER                               │
│  Spot Price · Option Chain · Open Interest · Volume · Headlines · Calendar   │
└──────────────────────────────────────────────────────────────────────────────┘
        │          │          │          │          │          │
        ▼          ▼          ▼          ▼          ▼          ▼

┌──────────────┐┌──────────┐┌──────────┐┌──────────┐┌──────────┐┌──────────────────┐
│ Gamma Flip   │ GEX      │ Vol      │ Flow     │ Dealer   │ Macro / Sched.   │
│ gamma_       │ gamma_   │ regime   │ imbal.   │ inventory│ Events / News    │
│ regime       │ exposure │ vol_     │ flow_    │ dealer_  │ macro_regime     │
│ (3 vals)     │ (2 vals) │ regime   │ signal   │ position │ event_window_    │
│              │          │ (3 vals) │ (3 vals) │ (2 vals) │ status (7 vals)  │
└──────────────┘└──────────┘└──────────┘└──────────┘└──────────┘└──────────────────┘
      │           │          │          │          │            │
      ▼           ▼          ▼          ▼          ▼            ▼

┌──────────────────────────────────────┐  ┌──────────────────────────────────────┐
│       ANALYTICS LAYER (derived)      │  │          RISK / OVERLAY LAYER        │
│ market_gamma_map → gamma_regime (4)  │  │ global_risk_state (5 vals)           │
│ vol_surface → vol_regime (4)         │  │ global_risk_action (4 vals)          │
│ smart_money → smart_flow (4)         │  │ global_risk_level (3 vals)           │
│ hedging_sim → hedging_bias (5)       │  │ squeeze_risk_state (4 vals)          │
│ gamma_shift → shift_signal (3→3)     │  │ directional_convexity_state (4 vals) │
│ liquidity → liq_signal (2+2+2)       │  │ dealer_flow_state (5 vals)           │
└──────────────────────────────────────┘  └──────────────────────────────────────┘
                │                                        │
                └────────────────────────────────────────┘
                                   ▼
                      ┌──────────────────────────────┐
                      │     SIGNAL ENGINE LAYER      │
                      │ data_quality_status (4 vals) │
                      │ confirmation_status (4 vals) │
                      │ direction (CALL/PUT/None)    │
                      │ signal_regime (6 vals)       │
                      │ execution_regime (5 vals)    │
                      │ trade_status (6 vals)        │
                      │ regime_fingerprint (hash)    │
                      └──────────────────────────────┘
```

Category A — Macro Regime

Source: `macro/macro_news_aggregator.py`
Field name: `macro_regime`
Downstream consumers: `risk/global_risk_regime.py`, `engine/signal_engine.py`

#	Label	Decision Rule	Interpretation
1	`EVENT_LOCKDOWN`	`event_lockdown_flag == True` (hard override from event window)	All trading blocked; scheduled macro event imminent or live

#	Label	Decision Rule	Interpretation
2	`RISK_OFF`	Any of: `sentiment ≤ −18` | `global_bias ≤ −0.22` | `india_bias ≤ −0.22` | `vol_shock ≥ 65`	Bearish macro conditions; position sizing reduced
3	`RISK_ON`	`sentiment ≥ 18` AND `global_bias ≥ 0.22` AND `vol_shock < 55`	Bullish macro conditions; full sizing allowed
4	`MACRO_NEUTRAL`	Fallback when none of the above triggers	No strong macro directional bias

Priority order: EVENT_LOCKDOWN → RISK_OFF → RISK_ON → MACRO_NEUTRAL (first match wins)

Category B — Event Window Status

Source: `macro/scheduled_event_risk.py`
Field name: `event_window_status`
Downstream consumers: `macro/macro_news_aggregator.py` → `macro_regime`

#	Label	Decision Rule	Interpretation
5	`CLEAR`	No active event in the schedule	No event risk; risk = 0
6	`PRE_EVENT_WATCH`	`lockdown_min < minutes_until_event ≤ warning_min` (warning = 180 min)	Event approaching; heightened awareness, no lockdown
7	`PRE_EVENT_LOCKDOWN`	`minutes_until_event ≤ lockdown_min` (lockdown = 30 min)	Event imminent; `event_lockdown_flag = True`
8	`LIVE_EVENT`	`0 ≤ minutes_since_event ≤ event_duration` (duration = 10 min)	Event in progress; `event_lockdown_flag = True`
9	`POST_EVENT_COOLDOWN`	`event_duration < minutes_since_event ≤ cooldown_min` (cooldown = 30 min)	Event concluded; residual volatility possible
10	`NO_EVENT_DATA`	Schedule file missing or empty	Treat as no events; risk = 0
11	`EVENT_FILTER_DISABLED`	Event filtering feature disabled in policy	Feature disabled; risk = 0

Category C — Event Severity Level

Source: `config/event_window_policy.py`
Field name: `severity`
Downstream consumers: `macro/scheduled_event_risk.py` → base risk calculation

#	Label	Base Risk Score	Interpretation
12	`CRITICAL`	95	RBI policy, GDP print, major global event
13	`MAJOR`	80	CPI/WPI release, FOMC, PMI
14	`MEDIUM`	55	Trade balance, industrial production
15	`MINOR`	30	Secondary indicators, regional data

Category D — Global Risk State

Source: `risk/global_risk_regime.py`
Field name: `global_risk_state`
Inputs: `volatility_explosion_probability`, `event_lockdown_flag`, `macro_event_risk_norm`, `regime_score`, `macro_regime`
Downstream consumers: `engine/signal_engine.py` → trade decision modifiers

#	Label	Decision Rule	Adjustment	Size Cap
16	`VOL_SHOCK`	`volatility_explosion_probability > 0.7`	-6	0.35
17	`EVENT_LOCKDOWN`	`event_lockdown_flag == True` OR `macro_event_risk_norm > 0.7`	-4	0.0 (if lockdown) / 0.65
18	`RISK_OFF`	`regime_score > 0.6` OR `macro_regime == "RISK_OFF"`	-4	0.65
19	`RISK_ON`	`regime_score < -0.3` AND `neutral_fallback == False`	0	1.0
20	`GLOBAL_NEUTRAL`	Fallback (none of the above)	0	1.0

Priority order: VOL_SHOCK → EVENT_LOCKDOWN → RISK_OFF → RISK_ON → GLOBAL_NEUTRAL

Category E — Global Risk Action

Source: `risk/global_risk_layer.py`
Field name: `global_risk_action`
Downstream consumers: `engine/signal_engine.py` → trade admission gate

#	Label	Decision Rule	Interpretation
21	`BLOCK`	`data_fatal` OR `event_lockdown` OR `global_risk_veto`	Trade rejected; hard block
22	`WATCHLIST`	Overnight gap / confirmation / strength / data caution flags	Trade demoted to watchlist

#	Label	Decision Rule	Interpretation
23	`REDUCE`	`size_cap < 1.0` and no block triggers	Trade allowed at reduced size
24	`ALLOW`	All checks passed	Full trade allowed

Category F — Global Risk Level

Source: `risk/global_risk_layer.py`
Field name: `global_risk_level`
Input: `global_risk_score` (0–100)

#	Label	Threshold	Interpretation
25	`LOW`	Score 0–34	Benign risk environment
26	`MEDIUM`	Score 35–59	Moderate risk; caution warranted
27	`HIGH`	Score 60–100	Elevated risk; defensive posture

Category G — Gamma Regime

G.1 — Gamma Flip Classification

Source: `analytics/gamma_flip.py`
Field name: `gamma_regime`

#	Label	Decision Rule	Interpretation
28	`POSITIVE_GAMMA`	`spot > gamma_flip_level`	Dealers long gamma; market self-stabilizing
29	`NEGATIVE_GAMMA`	`spot ≤ gamma_flip_level`	Dealers short gamma; market self-amplifying
30	`UNKNOWN`	Flip level cannot be computed	Insufficient data

G.2 — Market Gamma Map Classification

Source: `analytics/market_gamma_map.py`
Field name: `gamma_regime`

#	Label	Decision Rule	Interpretation
31	`POSITIVE_GAMMA`	`total_gex > 0` AND `abs(total_gex) > gross_gex × 0.05`	Net positive dealer gamma exposure
32	`NEGATIVE_GAMMA`	`total_gex < 0` AND `abs(total_gex) > gross_gex × 0.05`	Net negative dealer gamma exposure

#	Label	Decision Rule	Interpretation
33	`NEUTRAL_GAMMA`	`abs(total_gex) ≤ gross_gex × 0.05`	Balanced gamma; neither positive nor negative
34	`UNKNOWN`	GEX computation fails	Insufficient data

Category H — Gamma Exposure Signal

Source: `analytics/gamma_exposure.py`
Field name: `gamma_exposure_signal`

#	Label	Decision Rule	Interpretation
35	`LONG_GAMMA`	`total_gamma > 0`	Net long gamma in the chain
36	`SHORT_GAMMA`	`total_gamma ≤ 0`	Net short gamma in the chain

Category I — Volatility Regime

I.1 — Realized Volatility Regime

Source: `analytics/volatility_regime.py`
Field name: `volatility_regime`
Input: Intraday returns standard deviation
Config: `config/analytics_feature_policy.py` → `VolatilityRegimePolicyConfig`

#	Label	Decision Rule	Interpretation
37	`LOW_VOL`	`realized_vol < 0.01`	Subdued volatility; potential compression
38	`NORMAL_VOL`	`0.01 ≤ realized_vol < 0.03`	Typical volatility conditions
39	`VOL_EXPANSION`	`realized_vol ≥ 0.03`	Elevated volatility; regime shift possible

I.2 — Implied Volatility Regime (from Surface)

Source: `analytics/volatility_surface.py`
Field name: `vol_regime`

#	Label	Decision Rule	Interpretation
40	`HIGH_VOL`	`implied_vol > 25`	Elevated IV; premium rich
41	`NORMAL_VOL`	`15 ≤ implied_vol ≤ 25`	Typical IV conditions
42	`LOW_VOL`	`implied_vol < 15`	Compressed IV; potential expansion ahead
43	`UNKNOWN`	IV computation fails	Insufficient data

Category J — Intraday Gamma Shift & Volatility Implication

Source: `analytics/intraday_gamma_shift.py`
Field name: `gamma_shift` → mapped to `vol_implication`

Gamma Shift Signal

#	Label	Decision Rule	Interpretation
44	`NO_SHIFT`	`abs(gamma_change) < noise_threshold` (5%)	No meaningful gamma movement
45	`GAMMA_INCREASE`	`gamma_change ≥ noise_threshold`	Gamma exposure increasing
46	`GAMMA_DECREASE`	`gamma_change ≤ -noise_threshold`	Gamma exposure decreasing

Volatility Implication (Derived from Gamma Shift)

#	Label	Mapped From	Interpretation
47	`VOL_EXPANSION`	`GAMMA_DECREASE`	Less gamma hedging → more volatility
48	`VOL_SUPPRESSION`	`GAMMA_INCREASE`	More gamma hedging → less volatility
49	`NEUTRAL`	`NO_SHIFT`	No volatility implication

Category K — Flow Signal

Source: `analytics/options_flow_imbalance.py`
Field name: `flow_signal`
Input: Call/put volume imbalance ratio
Config: `config/analytics_feature_policy.py` → `FlowImbalancePolicyConfig`

#	Label	Decision Rule	Interpretation
50	`BULLISH_FLOW`	`imbalance_ratio ≥ 1.20`	Call-heavy flow dominance
51	`BEARISH_FLOW`	`imbalance_ratio ≤ 0.83`	Put-heavy flow dominance
52	`NEUTRAL_FLOW`	`0.83 < imbalance_ratio < 1.20`	Balanced call/put activity

Category L — Smart Money Flow

Source: `analytics/smart_money_flow.py`
Field name: `smart_money_signal`

Input: Large-block/unusual volume ratio
Config: `config/analytics_feature_policy.py` → `SmartMoneyFlowPolicyConfig`

#	Label	Decision Rule	Interpretation
53	`BULLISH_FLOW`	`smart_ratio ≥ 1.15`	Institutional call-side accumulation
54	`BEARISH_FLOW`	`smart_ratio ≤ 0.87`	Institutional put-side accumulation
55	`MIXED_FLOW`	`0.87 < smart_ratio < 1.15`	Ambiguous institutional activity
56	`NO_FLOW`	No qualifying large-block activity detected	No institutional signal

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Category M — Dealer Positioning

Source: `analytics/dealer_inventory.py`
Field name: `dealer_position`

#	Label	Decision Rule	Interpretation
57	`Long Gamma`	Net dealer gamma > 0	Dealers will hedge against moves (stabilizing)
58	`Short Gamma`	Net dealer gamma ≤ 0	Dealers will hedge with moves (amplifying)

Category N — Dealer Hedging Bias

N.1 — Hedging Flow Direction

Source: `analytics/dealer_hedging_flow.py`
Field name: `dealer_hedging_flow`

#	Label	Decision Rule	Interpretation
59	`BUY_FUTURES`	Net dealer hedging requires buying	Upside pressure from hedging
60	`SELL_FUTURES`	Net dealer hedging requires selling	Downside pressure from hedging

N.2 — Hedging Bias (from Simulator)

Source: `analytics/dealer_hedging_simulator.py`
Field name: `dealer_hedging_bias`
Input: Spot vs key gamma levels (2% tolerance threshold)

#	Label	Decision Rule	Interpretation
61	`PINNING`	Spot within 2% of major gamma level; balanced gamma	Price pinned by hedging flows
62	`UPSIDE_PINNING`	Spot near positive gamma wall above	Resistance from dealer hedging

#	Label	Decision Rule	Interpretation
63	`DOWNSIDE_PINNING`	Spot near negative gamma wall below	Support from dealer hedging
64	`UPSIDE_ACCELERATION`	Spot breaking above in negative gamma zone	Dealer hedging amplifies upside
65	`DOWNSIDE_ACCELERATION`	Spot breaking below in negative gamma zone	Dealer hedging amplifies downside

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Category O — Liquidity Signal

O.1 — Liquidity Heatmap

Source: `analytics/liquidity_heatmap.py`
Field name: `liquidity_signal`

#	Label	Decision Rule	Interpretation
66	`STRONG_LEVEL_NEAR`	High OI concentration within 50 points of spot	Strong support/resistance nearby
67	`LEVEL_FAR`	No significant OI concentration within 50 points	No proximate liquidity wall

O.2 — Liquidity Vacuum

Source: `analytics/liquidity_vacuum.py`
Field name: `vacuum_direction`

#	Label	Decision Rule	Interpretation
68	`BREAKOUT_ZONE`	OI in adjacent strikes < 25% of previous level	Void in liquidity; sharp move risk
69	`NORMAL`	Adequate OI in adjacent strikes	No vacuum; orderly market

O.3 — Liquidity Void

Source: `analytics/liquidity_void.py`
Field name: `liquidity_void_signal`

#	Label	Decision Rule	Interpretation
70	`VOID_NEAR`	OI < 50 threshold within 50 points of spot	Nearby liquidity gap; slippage risk
71	`VOID_FAR`	OI < 50 threshold exists but beyond 50 points	Distant void; less immediate risk

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Category P — Squeeze Risk State

Source: `risk/gamma_vol_acceleration_regime.py` → `_classify_squeeze_risk_state()`
Field name: `squeeze_risk_state`
Input: `gamma_vol_acceleration_score` (0–100)
Config: `config/gamma_vol_acceleration_policy.py`

#	Label	Threshold	Interpretation
72	`EXTREME_ACCELERATION_RISK`	Score ≥ 78	Gamma squeeze extremely likely; maximum caution
73	`HIGH_ACCELERATION_RISK`	Score ≥ 60	High squeeze probability; reduced sizing
74	`MODERATE_ACCELERATION_RISK`	Score ≥ 40	Moderate squeeze risk; monitor closely
75	`LOW_ACCELERATION_RISK`	Score < 40	Low squeeze risk; normal operations

Category Q — Directional Convexity State

Source: `risk/gamma_vol_acceleration_regime.py` → `_directional_state()`
Field name: `directional_convexity_state`
Inputs: `gamma_vol_acceleration_score`, `upside_risk` (0–1), `downside_risk` (0–1)
Config thresholds: `directional_edge = 0.58`, `two_sided_edge = 0.48`, `two_sided_balance_tolerance = 0.12`

#	Label	Decision Rule	Interpretation
76	`UPSIDE_SQUEEZE_RISK`	`upside ≥ 0.58` AND `upside > downside`	Directional gamma squeeze risk to the upside
77	`DOWNSIDE_AIRPOCKET_RISK`	`downside ≥ 0.58` AND `downside > upside`	Directional air-pocket risk to the downside
78	`TWO_SIDED_VOLATILITY_RISK`	Both `upside ≥ 0.48` AND `downside ≥ 0.48` AND `abs(upside - downside) ≤ 0.12`	Bidirectional volatility risk
79	`NO_CONVEXITY_EDGE`	Score < 25 AND `max(upside, downside) < 0.25`; or fallback	No significant convexity edge

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Category R — Dealer Flow State

Source: `risk/dealer_hedging_pressure_regime.py` → `_state()`
Field name: `dealer_flow_state`
Inputs: `upside_pressure` (0–1), `downside_pressure` (0–1), `pinning_pressure` (0–1)
Config thresholds: `upside/downside = 0.60`, `pinning = 0.62`, `two_sided = 0.48`, `balance_tolerance = 0.12`

#	Label	Decision Rule	Interpretation
80	`UPSIDE_HEDGING_ACCELERATION`	`upside ≥ 0.60` AND `upside > downside + 0.12`	Dealer hedging drives upside momentum
81	`DOWNSIDE_HEDGING_ACCELERATION`	`downside ≥ 0.60` AND `downside > upside + 0.12`	Dealer hedging drives downside momentum
82	`PINNING_DOMINANT`	`pinning ≥ 0.62` AND `pinning > max(upside, downside) + 0.08`	Dealer hedging pins price to gamma level
83	`TWO_SIDED_INSTABILITY`	Both `upside ≥ 0.48` AND `downside ≥ 0.48` ; or near-balanced pressures ≥ 0.38	Unstable hedging; whipsaw risk
84	`HEDGING_NEUTRAL`	Fallback (none of the above)	No dominant hedging pressure

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Category S — Data Quality Status

Source: `engine/trading_support/signal_state.py`
Field name: `data_quality_status`
Input: `data_quality_score` (starts at 100, penalties applied)
Penalties: Invalid spot (-45), stale spot (-10), invalid chain (-45), stale chain (-10), provider WEAK (-18), provider CAUTION (-8), missing analytics (variable)

#	Label	Threshold	Interpretation
85	`STRONG`	Score ≥ 85	All data sources healthy
86	`GOOD`	Score ≥ 70	Minor data gaps; signals reliable
87	`CAUTION`	Score ≥ 55	Significant data issues; signals degraded
88	`WEAK`	Score < 55	Critical data missing; signals unreliable

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Category T — Provider Health Status

Source: `data/option_chain_validation.py`
Field name: `provider_health.summary_status`
Downstream consumers: Data quality score (Category S)

#	Label	Decision Rule	Penalty
89	`WEAK`	Any individual provider rated WEAK	-18 to data quality
90	`CAUTION`	Any individual provider rated CAUTION (none WEAK)	-8 to data quality
91	`GOOD`	All providers healthy (implicit default)	No penalty

Category U — Confirmation Status

Source: `strategy/confirmation_filters.py`
Field name: `confirmation_status`
Input: Composite score from 8 confirmation voters
Config: `config/signal_policy.py` → `CONFIRMATION_FILTER_CONFIG`

#	Label	Threshold	Interpretation
92	`STRONG_CONFIRMATION`	Score ≥ 6	Multiple confirming signals aligned
93	`CONFIRMED`	Score ≥ 2	Adequate confirmation for trade
94	`MIXED`	Score > -3	Partial confirmation; watchlist candidate
95	`CONFLICT`	Score ≤ -3	Conflicting signals; trade suppressed

Confirmation Voters (8 components)

Voter	Field	Logic
Open Alignment	`open_alignment_score`	Spot direction vs session open
Prev Close Alignment	`prev_close_alignment_score`	Spot direction vs previous close
Range Expansion	`range_expansion_score`	Intraday range vs microstructure thresholds
Flow Confirmation	`flow_confirmation_score`	Flow signal polarity match
Hedging Confirmation	`hedging_confirmation_score`	Hedging bias alignment (strong weight)
Gamma Event Confirmation	`gamma_event_confirmation_score`	Gamma squeeze event boost
Move Probability Confirmation	`move_probability_confirmation_score`	Hybrid move probability thresholds
Flip Alignment	`flip_alignment_score`	Spot vs gamma flip level alignment

Category V — Trade Direction

Source: `engine/trading_support/signal_state.py` → `decide_direction()`
Field names: `direction`, `direction_source`

#	Label	Decision Rule	Interpretation
96	`CALL`	`bullish_score ≥ 1.75` AND margin ≥ 0.7 AND bullish > bearish	Bullish directional trade
97	`PUT`	`bearish_score ≥ 1.75` AND margin ≥ 0.7 AND bearish > bullish	Bearish directional trade

#	Label	Decision Rule	Interpretation
98	`None`	Neither side meets minimum score/margin thresholds	No directional conviction

Direction Vote Weights

Voter	Weight	Source Signal
FLOW	1.20	`final_flow_signal` polarity
HEDGING_BIAS	1.10	`dealer_hedging_bias` direction
GAMMA_SQUEEZE	0.90	Squeeze conditions detected
GAMMA_FLIP	0.85	Spot vs flip level position
DEALER_VOL	0.80	Dealer volatility positioning
VANNA	0.45	Vanna exposure signal
CHARM	0.45	Charm decay signal
BACKTEST_FALLBACK	0.60	Historical backtest bias

`direction_source` format: Concatenation of winning voter names, e.g., `"FLOW+HEDGING_BIAS+GAMMA_SQUEEZE"`

Category W — Signal Regime

Source: `engine/trading_support/signal_state.py` → `classify_signal_regime()`
Field name: `signal_regime`
Config thresholds: `expansion_bias = 75`, `directional_bias = 55`

#	Label	Decision Rule	Interpretation
99	`LOCKDOWN`	`event_lockdown_flag == True`	All signals suppressed; macro event active
100	`EXPANSION_BIAS`	`strength ≥ 75` AND directional flow AND unstable gamma AND confirmation ∈ {STRONG, CONFIRMED}	Maximum conviction setup; gamma squeeze likely
101	`DIRECTIONAL_BIAS`	`strength ≥ 55` AND directional flow	Clear directional edge detected
102	`CONFLICTED`	`data_quality ∈ {CAUTION, WEAK}` OR `confirmation == CONFLICT`	Signals unreliable or contradictory
103	`BALANCED`	Falls through all checks (with valid direction)	Moderate conviction; standard setup
104	`NEUTRAL`	`direction is None`	No directional bias detected

Category X — Execution Regime

Source: `engine/trading_support/signal_state.py` → `classify_execution_regime()`
Field name: `execution_regime`
Config thresholds: `reduced_size_multiplier = 1.0`, `observe_data_quality = 70`

#	Label	Decision Rule	Interpretation
105	`BLOCKED`	`trade_status ∈ {DATA_INVALID, EVENT_LOCKDOWN, NO_TRADE, BUDGET_FAIL}`	Trade execution prevented
106	`RISK_REDUCED`	`trade_status == TRADE` AND `size_multiplier < 1.0`	Trade allowed at reduced position size
107	`ACTIVE`	`trade_status == TRADE` (full size)	Full execution authorized
108	`OBSERVE`	`signal_regime == CONFLICTED` OR `data_quality_score < 70`	Monitor only; no execution
109	`SETUP`	Fallback (watchlist or near-actionable)	Setup forming; awaiting confirmation

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Category Y — Trade Status

Source: `engine/signal_engine.py`
Field name: `trade_status`
**Final trade admission decision

#	Label	Decision Rule	Interpretation
110	`TRADE`	All gates passed; direction confirmed; strength above threshold	Execute trade
111	`WATCHLIST`	Partial conditions met; insufficient confirmation or strength	Monitor for entry
112	`NO_TRADE`	Signal too weak or conditions unfavorable	No opportunity
113	`DATA_INVALID`	Critical data missing or corrupt	Cannot evaluate
114	`NO_SIGNAL`	No meaningful signal generated	Flat market conditions
115	`BUDGET_FAIL`	Risk budget exhausted	Position limits reached

Category Z — Capture Policy

Source: `research/signal_evaluation/policy.py`
Field name: `capture_policy`
Used by: Signal evaluation framework for research capture filtering

#	Label	Captures	Interpretation
116	`TRADE_ONLY`	`trade_status == "TRADE"`	Capture only executed trades
117	`ACTIONABLE`	`trade_status ∈ {"TRADE", "WATCHLIST"}`	Capture trades and watchlist signals
118	`ALL_SIGNALS`	All non-empty `trade_status`	Capture everything for research

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Category AA — Composite Risk Metrics

These are continuous (0–1) scores rather than discrete labels. They feed into the categorical classifications above.

AA.1 — Risk-Off Intensity

Source: `risk/global_risk_regime.py`
Field name: `risk_off_intensity`
Range: 0.0 – 1.0

Formula:

```
risk_off_intensity = 0.28 × volatility_component
                   + 0.20 × us_equity_component
                   + 0.12 × rates_component
                   + 0.12 × fx_component
                   + 0.16 × commodity_component
                   + 0.12 × macro_event_component
```

AA.2 — Volatility Explosion Probability

Source: `risk/global_risk_regime.py`
Field name: `volatility_explosion_probability`
Range: 0.0 – 1.0
Critical threshold: > 0.7 triggers `VOL_SHOCK` state

Formula:

```
volatility_explosion_probability = compression_score × (vol_shock_norm + macro_event_norm)
```

AA.3 — Volatility Compression Score

Source: `risk/global_risk_regime.py`
Field name: `volatility_compression_score`
Range: 0.0 – 1.0
Input: Ratio of 5-day to 30-day realized volatility

Compression Level	Ratio Range	Score
Extreme	< 0.45	Near 1.0
Medium	0.45 – 0.60	0.5 – 0.8
Low	0.60 – 0.75	0.2 – 0.5
None	≥ 0.75	Near 0.0

Category AB — Regime Fingerprint

Source: `research/signal_evaluation/evaluator.py` → `build_regime_fingerprint()`
Field names: `regime_fingerprint`, `regime_fingerprint_id`

A deterministic hash encoding the full regime state for signal evaluation and deduplication.

Components

```
signal_regime | macro_regime | gamma_regime | spot_vs_flip | flow
dealer_pos | hedging | vol | vacuum | confirm | dataq | provider
```

Format: `"signal_regime=EXPANSION_BIAS|macro_regime=RISK_ON|gamma_regime=NEGATIVE_GAMMA|..."`
`regime_fingerprint_id`: First 16 characters of SHA-256 hash of the fingerprint string.

Final Summary

Statistics

Metric	Count
Total distinct symbolic labels	118
Categorical classifications	28 categories
Composite continuous metrics	3
Confirmation voters	8
Direction voters	8
Layers traversed	5 (Data → Analytics → Risk/Overlay → Signal → Trade)

Ambiguities and Naming Inconsistencies

Issue	Details	Suggestion
Duplicate `gamma_regime` field	`gamma_flip.py` (3 values) and `market_gamma_map.py` (4 values) both produce `gamma_regime`	Rename one to `gamma_flip_regime` vs `gamma_map_regime`
Overlapping `volatility_regime`	`volatility_regime.py` (realized vol) and `volatility_surface.py` (implied vol) use similar names	Rename to `realized_vol_regime` vs `implied_vol_regime`
Mixed flow label sets	`options_flow_imbalance.py` uses `NEUTRAL_FLOW` while `smart_money_flow.py` uses `MIXED_FLOW` and `NO_FLOW`	Standardize: use `NEUTRAL_FLOW` / `NO_SIGNAL` consistently
Dealer position strings	`dealer_inventory.py` uses `"Long Gamma"` / `"Short Gamma"` (spaces, title case) while all other labels use `UPPER_SNAKE_CASE`	Standardize to `LONG_GAMMA` / `SHORT_GAMMA`
`EVENT_LOCKDOWN` collision	Both `macro_regime` and `global_risk_state` define `EVENT_LOCKDOWN` with slightly different triggers	Prefix: `MACRO_EVENT_LOCKDOWN` vs `GLOBAL_EVENT_LOCKDOWN`
`RISK_OFF` collision	Both `macro_regime` and `global_risk_state` define `RISK_OFF`	Prefix consistently or keep clear which layer owns the label

Standardization Recommendations

1. **Adopt UPPER_SNAKE_CASE universally** for all state labels (fix `"Long Gamma"` / `"Short Gamma"` in `dealer_inventory.py`)
2. **Prefix ambiguous labels** with their layer: `MACRO_`, `GLOBAL_`, `GAMMA_FLIP_`, `GAMMA_MAP_`
3. **Unify flow signal taxonomy** — choose one set of labels for bullish/bearish/neutral/no-signal across flow modules
4. **Document all threshold values in a central config registry** to avoid magic numbers scattered across modules
5. **Add `UNKNOWN` fallback** to every classifier that operates on potentially missing data (some classifiers already do this; make it universal)